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Proton beam radiotherapy of uveal melanoma: Italian patients treated in Nice, France

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PURPOSE. *To evaluate the results of 15 years of experience with proton beam radiotherapy in the treatment of intraocular melanoma, and to determine univariate and multivariate risk factors for local failure, eye retention, and survival.*

METHODS. *A total of 368 cases of intraocular melanoma were treated with proton beam radiotherapy at Centre Lacassagne Cyclotron Biomedical of Nice, France, between 1991 and 2006. Actuarial methods were used to evaluate rate of local tumor control, eye retention, and survival after proton beam radiotherapy. Cox regression models were extracted to evaluate univariate risk factors, while regularized least squares algorithm was used to have a multivariate classification model to better discriminate risk factors.*

RESULTS. *Tumor relapse occurred in 8.4% of the eyes, with a median recurrence time of 46 months. Enucleation was performed on 11.7% of the eyes after a median time of 49 months following proton beam; out of these, 29 eyes were enucleated due to relapse and 16 due to other causes. The univariate regression analysis identified tumor height and diameter as primary risk factors for enucleation. Regularized least squares analysis demonstrated the higher effectiveness of a multivariate model of five risk factors (macula distance, optic disc distance, tumor height, maximum diameter, and age) in discriminating relapsed vs nonrelapsed patients.*

CONCLUSIONS. *This data set, which is the largest in Italy with relatively long-term follow-up, demonstrates that a high rate of tumor control, survival, and eye retention were achieved after proton beam irradiation, as in other series. (Eur J Ophthalmol 2009; 19: 654-60)*

KEY WORDS. *Intraocular melanoma, Proton beam, Radiotherapy*

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INTRODUCTION

Uveal melanoma is the most common eye primary tumor (1). The annual incidence in the United States is six cases per million; in Italy, 350–400 new cases are estimated per year. Radiotherapy (external beam with charged particle–proton/helium, or episcleral plaque therapy) (2-4) and surgery associated with radiotherapy (5) are the preferred treatments for most patients with this tumor. Enucleation is performed only in cases where tumor and eye conditions are not convenient for radiotherapy (volume of the tumor not suitable for radiotherapy, low vision associated with total or subtotal retinal detachment, special cases). With radiotherapy eye salvage is achieved, and, particularly for cases in which the tumor is located away or far from the posterior (optic disc and macula area), useful vision is retained after treatment. Previous reports confirm high rates of

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local tumor control with 5 years control rates superior to 95% for tumors treated with proton beam. Survival rate is not to be compromised with conservative therapy compared to enucleation, but few investigators have reported an increased risk of death from metastasis when the treatment has failed to control local tumor after radiotherapy (6). In this study, we evaluated the results of local tumor control, eye retention, and survival after proton beam radiotherapy performed at the Ocular Oncology Service, Genoa, Italy.

METHODS

We evaluated local tumor control, eye retention, and survival in a cohort of 368 patients with intraocular melanoma treated with proton therapy at the Ocular Oncology Service between 1991 and 2006 and followed prospectively through December 2007. Patients who received previous therapy or adjuvant therapy (transpupillary thermotherapy) after proton irradiation were excluded. Additionally, patients with bilateral or iris tumor and patients diagnosed with metastasis at time of presentation were excluded from analysis.

The median follow-up was 3.9 years. The initial diagnosis and follow-up were done at the Ocular Oncology Service in Genoa. Tumor characteristics determined during the initial ophthalmologic examination including tumor size, shape, and location were recorded in a dedicated database. Demographic and patient characteristics, including age and gender, were recorded. Pretreatment work-up including liver function and liver ultrasound was performed every 6 months and chest x-rays every 12 months to rule out systemic metastasis. The proton beam treatment was done at the Centre Antoine Lacassagne Cyclotron Biomedical, Nice, France. The standard protocols require delivery of 60 Gy equivalent photons in 4 fractions in 4 consecutive days. Clips insertion on the sclera for tumor localization was performed by the same surgeon in each patient; the surgeon collaborated with the radiotherapist in Nice in the elaboration of the treatment plan (7). Different technical systems (computed tomography scan for eye reconstruction, security margin varying from 1.5 to 2.5 mm, wedge filter, tumor or contralateral fixation) are currently used in the elaboration of the treatment plan looking at the different localization and size of the tumor, to have good eye retention and save as much vision as possible. Ocular outcomes, including tumor relapse and eye retention, were ascertained through December 2007. The patients returned to the Ocular Oncology Service in Genova for at least one follow-up and at least one time a year (8). Enucleation causes were divided into relapse and other (neovascular glaucoma, eye subatrophy, other). Local recurrence was documented by ophthalmologic examination, ultrasonography, and sequential fundus photography. Using Kaplan-Meier methods, we estimated annual incidence rates and cumulative rates after treatment, with corresponding 95% confidence intervals (CI). We calculated relative risk estimates using Cox proportional hazards regression, to determine statistically significant factors independently related to risk of tumor relapse, eye retention, and death. We also performed a multivariate data analysis by employing supervised learning techniques, in particular the algorithm known as regularized least squares (RLS) (9-11), in the classification of patients with and without tumor relapse. In the supervised learning or learning from examples setting, the goal is to infer an input-output relation given a finite number of input-output pairs, called training set. To achieve this, the learner has to generalize from the presented data to unseen situations (patients). In the RLS algorithm this is achieved through regularization. In our case, the inputs consisted of a panel of selected features characterizing each patient (macula distance, optic disc distance, tumor height, maximum diameter, and age), whereas the outputs were identified with a binary class label, relapsed (–1) or non-relapsed (1). Given a set of training patients, we inferred a multivariate classification pattern and tested it on an independent blind data set, hence determining its prediction accuracy; that is, the percentage of well classified test samples.

RESULTS

Approximately equal numbers of males and females were treated, with no predilection for either eye. This cohort was racially homogeneous (Caucasian subjects 100%). The average age at time of treatment was 62 years and the average tumor dimensions were 6.2 mm and 14.2 mm for maximum diameter and height, respectively. Tumors were predominantly located in the choroid and in 3.5% involved the ciliary body. No iris tumors were considered in our study. Only 4 (1.1%) eyes presented extrascleral extension of the tumor. Patients and tumor characteristics are listed in Tables I and II, respectively.

Tumor relapse occurred in 31 eyes (8.4% of the cohort). All these cases were documented by ultrasonography and sequential fundus photography. The median time of recurrence was 46 months (90% CI = 31, 65 months, min = 16, max = 422). A total of 43 eyes (11.7%) were enucleated after proton beam. Time of enucleation ranged between 10 and 186 months with a median of 49 months (90% CI = 33, 68 months). Of these, 29 eyes were enucleated due to relapse (67.4%) and 16 due to other causes (32.6%). A total of 39 (10.6%) patients presented metastasis and 18 patients (4.9%) died with metastasis after proton beam treatment. The median time both to develop metastasis and death were 63 months.

The survival rate observed during the follow-up after radiotherapy is 90% after 6 years and is reported in Figure 1. We performed the same evaluation for eye retention (Fig. 2) and local tumor control (Fig. 3), which are 82% after 6 years and 84% after 6 years, respectively. Statistically significant risk factors ($p<0.01$) for death, enucleation, and tumor relapse are listed in Table III: the most important ONES are tumor thickness and diameter.

The univariate regression analysis identified tumor height and tumor diameter as primary risk factors for both enucleation due to relapse and enucleation due to other causes (Tab. IV).

In the RLS analysis, we built a classification pattern discriminating patients with and without tumor relapse. For this analysis, we selected patients with robust prognosis; that is, patients with documented tumor relapse and patients relapse-free with more than 7 years of follow-up. Such shaving reduced the data set to a total of 75 patients, 31 and 44 respectively with and without tumor relapse. The classification algorithm has been employed within a strict validation protocol; that is, a two-layer 10-fold cross-validation loop for parameter setting and testing (12). In this case, we evaluated the good-

TABLE I - DEMOGRAPHIC DATA

Factor	Level	No. (%) of patients
Sex	Male	182 (49.5)
	Female	186 (50.5)
Age, yr (mean 62.3 yr)	50	67 (18.2)
	>50 3	01 (81.8)

TABLE II - TUMOR DATA

Factor	Level	No. (%) of patients
Tumor localization	Choroid	355 (96.5)
	Ciliary body	13 (3.5)
Tumor height, mm	5	163 (44.3)
(mean 6.2 mm)	>5	205 (55.7)
Extrascleral extension	Present	4 (1.1)
	Absent	364 (98.9)
Distance to macula,	2.5	146 (39.7)
mm (mean 5.5 mm)	>2.5	222 (60.3)
Distance to optic disc,	2.5	108 (29.3)
mm (mean 6.2 mm)	>2.5	260 (70.7)
Maximum diameter,	10	62 (16.8)
mm (mean 14.2 mm)	10–20	277 (75.3)
	>20	29 (7.9)

TABLE III - P VALUES OF COX PROPORTIONAL HAZARD MODEL

Death		Enucleation		Relapse	
Factor	p Value	Factor	p Value	Factor	p value
Macula distance	0.0049	Tumor height	0.0000	Tumor height	0.0000
Maximum diameter	0.0268	Maximum diameter	0.0001	Maximum diameter	0.0002
Optic disc distance	0.0774	Extrascleral extension	0.0975	Macula distance	0.1031
Tumor height	0.1780	Macula distance	0.1100	Extrascleral extension	0.2021
Gender	0.3299	Years (>50 yr)	0.2723	Location	0.2212
Location	0.5189	Location	0.3094	Years (>50 yr)	0.2919
Years (>50 yr)	0.6851	Optic disc distance	0.4307	Optic disc distance	0.5267
Extrascleral extension	0.7086	Gender	0.9536	Gender	0.8188

Factors are ordered with increasing p value. p Less than 0.01 indicates significant variables. Standard clinical thresholds are listed in Table II.

TABLE IV - P VALUES OF COX PROPORTIONAL HAZARD MODEL

Enucleation due to relapse		Enucleation due to other	
Factor	**p Value**	**Factor**	**p value**
Tumor height	0.0000	Tumor height	0.0225
Maximum diameter	0.0001	Maximum diameter	0.3082
Macula distance	0.0744	Macula distance	0.4988
Extrascleral extension	0.0923	Extrascleral extension	0.7812
Age	0.2352	Location	0.8380
Location	0.3169	Age	0.8541
Optic disc distance	0.4913	Optic disc distance	0.8761
Gender	0.8044	Gender	0.9754

Factors are ordered with increasing p value. p Lower than 0.01 indicates that the corresponding variable is significant. Standard clinical thresholds are listed in Table II.

TABLE V - PREDICTION ACCURACIES OF ORDINARY AND REGULARIZED LEAST SQUARE ESTIMATORS

Factor	Clinical accuracy	Ordinary least squares accuracy	RLS accuracy
Macula distance	59	56	—
Optic disc distance	49	57	—
Tumor height	67	63	—
Maximum diameter	66	71	—
Age	40	56	—
All risk factors	—	—	75

Values are percentages.

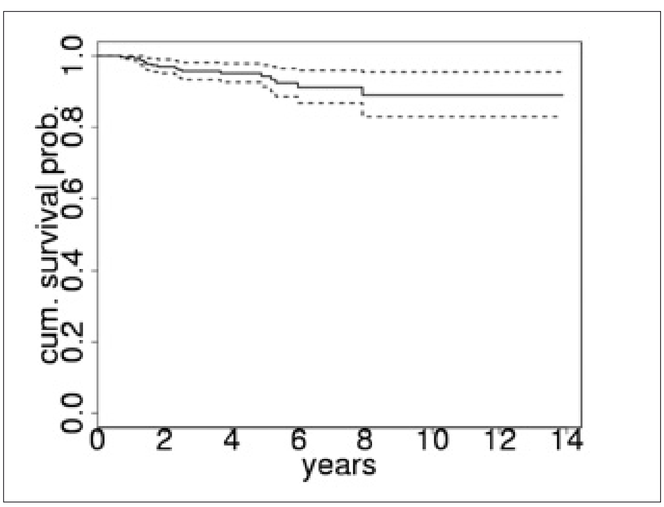

Fig. 1 - *Cumulative survival rate after proton therapy.*

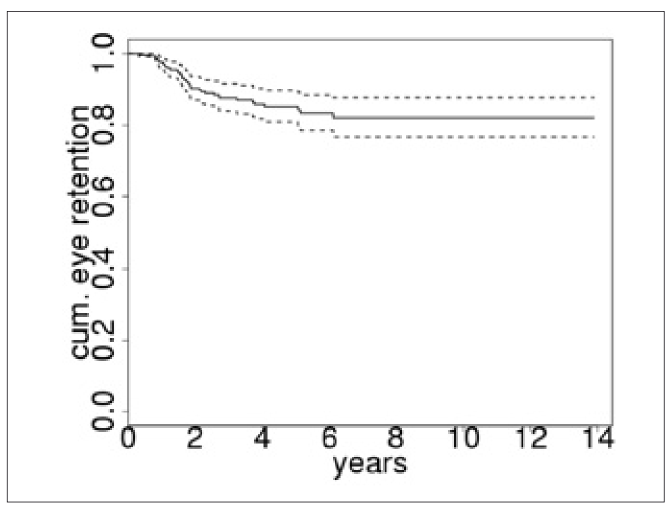

Fig. 2 - *Cumulative eye retention rate after proton therapy.*

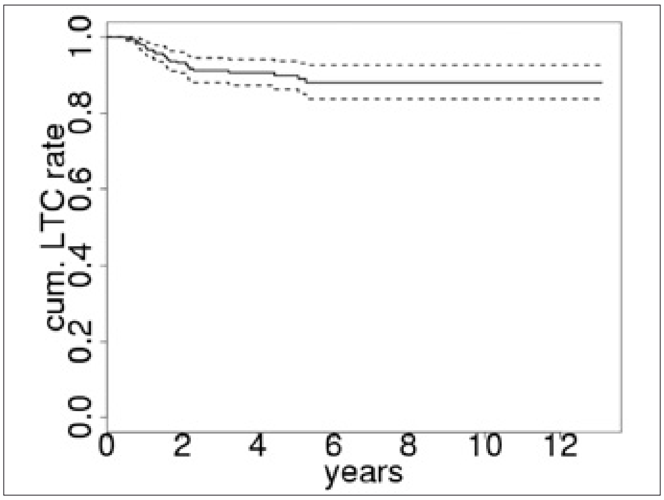

Fig. 3 - *Cumulative rate of local tumor control after proton therapy.*

ness of the model by means of the estimated performance on the test dataset, i.e., the cross-validation accuracy.

The RLS classification pattern associated to a panel of 5 risk factors (macula distance, optic disc distance, tumor height, maximum diameter, and age) demonstrated to be highly predictive, with 75% prediction accuracy. We then compared the accuracy of the multivariate RLS model to the performances associated to a set of univariate models based on each one of the risk factors, either using the classical clinical criterion based on thresholds or using a simple statistical learning algorithm, the ordinary least squares (OLS) (10), which is a not-regularized version of RLS. The choice of OLS in place of RLS is due to the simplicity of the model, based on just one variable at a time.

We reported the OLS prediction accuracies in Table V, together with the accuracies associated with the standard clinical thresholds, which correspond to the level employed in Cox proportional hazard model and reported in Table II. Note how significantly the prediction accuracy increased by using the RLS model. This can also be noted in Figure 4, where 5 plots are depicted. Each of them represents the patient distributions (relapsed displayed by stars and not relapsed showed in filled circles) with respect to each significant risk factor (lines 1 to 5) and the RLS estimator (bottom). In the first 5 plots the vertical lines represent the standard clinical thresholds, c.f. Table II, whereas in the lowest plot the line is the RLS classifier. The vertical lines divide the distributions into two classes and ideally, one should observe one whole group on the left side and the other one on the right. Note how the RLS classifier is the only one approximating this behavior.

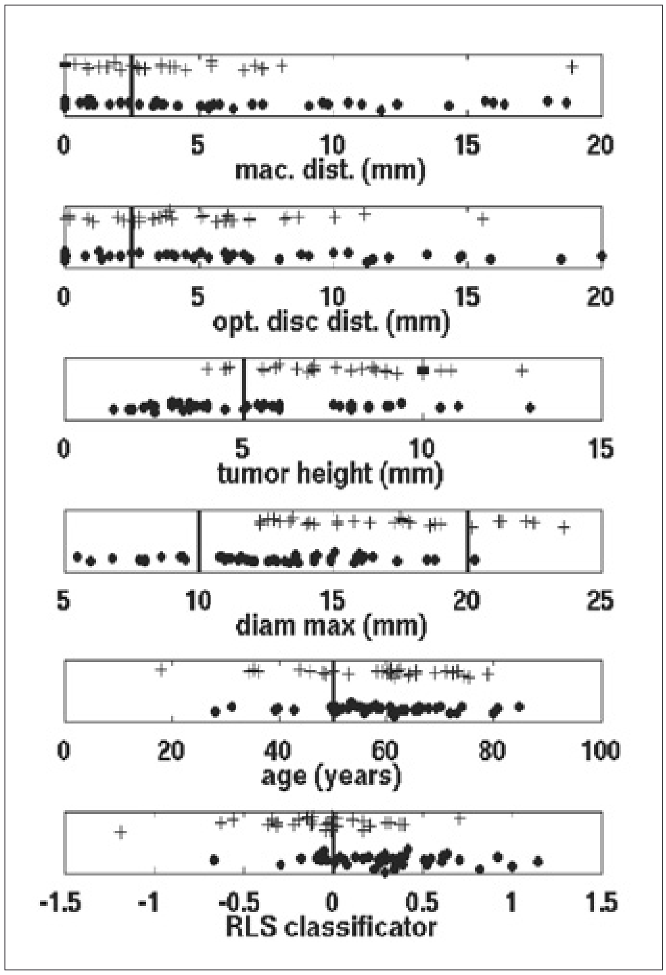

Fig. 4 - *Patient distributions with respect to significant factors and regularized least squares (RLS) estimator (bottom). Rows 1 to 5: graphical representation of 5 univariate classifiers; the horizontal lines represent the clinical thresholds as in Tables I and II. Row 6: graphical representation of RLS classifier; the horizontal line represents the classes boundary. The stars and filled circles represent relapsed and non-relapsed patients, respectively.*

DISCUSSION

Results of this study confirm previous studies and demonstrate that rates of relapse decrease with time after proton beam radiotherapy (13, 14). Local recurrence after radiotherapy is a prognostic indicator for tumor-related death (15-19). The cumulative rate of recurrence was approximately 10% 5 years after treatment. We found that patients with high and large tumors were at risk for local failure; this is in accord with the series of Courdi et al (20), which treated patients in the same center, at Lacassagne in Nice (20). This is confirmed by Egger et al (14) and Dendale et al (17); these authors reported local tumor control varying from 90% and 96% and considered tumor dimension the main cause for tumor failure. One possible explanation could be radiotherapy planning errors since the visualization of tumor margins by transillumination is more difficult when the tumor is large. Other possibilities are the evidence that tumor vascular networks are associated with an increased risk of metastasis and that these patterns are more often present in large tumors (21-23). Additionally, genetic aberration—monosomy 3, loss of chromosomal arms 6q, and polysomy of chromosome 8, associated with high risk of metastasis—is more frequent in these tumors (24-27). Large tumors are less radiosensitive than smaller tumors (28, 29). Further patients with large tumors and tumors involving the ciliary body are at increased risk for metastasis and death.

At the same time, metastasis and death have as primary risk the local relapse. This suggests that tumor characteristics (tumor thickness and tumor diameter) are the main indicators for risk of relapse, metastasis, and death and therefore are good guidelines for radiotherapy with proton beam in ocular melanoma.

Eye retention is another main goal of conservative treatment. The number of enucleated eyes in our study is 43, with an eye retention rate of 88.4%. Since 67.4% of the enucleations (29 eyes) were due to relapse, eye retention after proton beam radiotherapy is considered good in a cohort with medium and large tumors. This value of enucleation rate is confirmed by Damato et al (16) in a similar series of 349 tumors treated with proton beam and by Gragoudas and Lane (15) in over 2 decades of experience in proton beam treatment for ocular melanoma. Because Cox evaluation does not give an efficient prognostic model, it is necessary to use a different statistical system.

RLS proved to be an efficient tool for the statistical validation of our data. Indeed, RLS analysis demonstrated the higher effectiveness of a multivariate model of 5 risk factors (macula distance, optic disc distance, tumor height, maximum diameter, and age) in discriminating relapsed vs non-relapsed patients with respect to univariate clinical thresholds. Furthermore, the generalization property of RLS algorithm combined with an unbiased validation framework guarantees high prediction accuracy also in classifying independent patients (prospective study). The RLS proved to be an effective solution to provide a model able to predict the relapse status.

Overall, these data demonstrate that excellent local tumor control, survival, and eye retention rate were achieved following proton beam irradiation on our patients using a nonstandard procedure. This cohort is the largest in Italy, in collaboration with Centre Lacassagne in Nice, France, having relatively long-term follow-up. Future refinements in treatment planning (security margin, wedge filter, tumor or contralateral fixation), dosing, and delivery could be necessary to determine visual results (central vision, visual field, quality of vision) and complications after proton beam therapy in ocular melanoma.

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